

COGNITIVE STRATEGY TRANSFER

Why human management skills are the secret to better AI

- R01** Why do human strategies for managing complexity improve LLM performance?
- R02** When does that transfer hold, and where does it break down?
- Approach** Conceptual theory-building. A synthesis of prompting, cognitive science, education and management research, anchored in IS affordance theory.^{4,5}



THE PROBLEM

52 citations. 2 real.



The form of a citation, nothing behind it A citation we could verify

NOT A ONE-OFF. Lawyers sanctioned over six fake court cases (2023).¹ Deloitte part-refunds an A\$440,000 report with invented sources (2025).² KPMG: 40 of 45 reference titles fabricated (2025).³

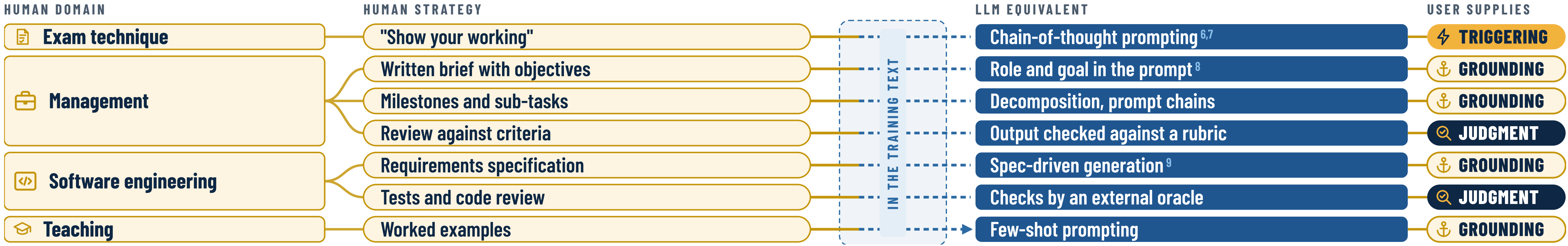
We asked a frontier model to draft a literature review. It gave us 52 confident citations. We could verify two.

Then we asked it to check its own references. It confirmed the fakes.

Every fake had the **form** of scholarship: authors, journal, volume, DOI. It lacked the **substance**, a real paper to point to. Only a check from **outside the model** could tell them apart.

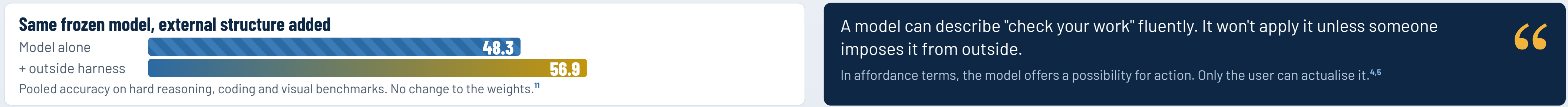
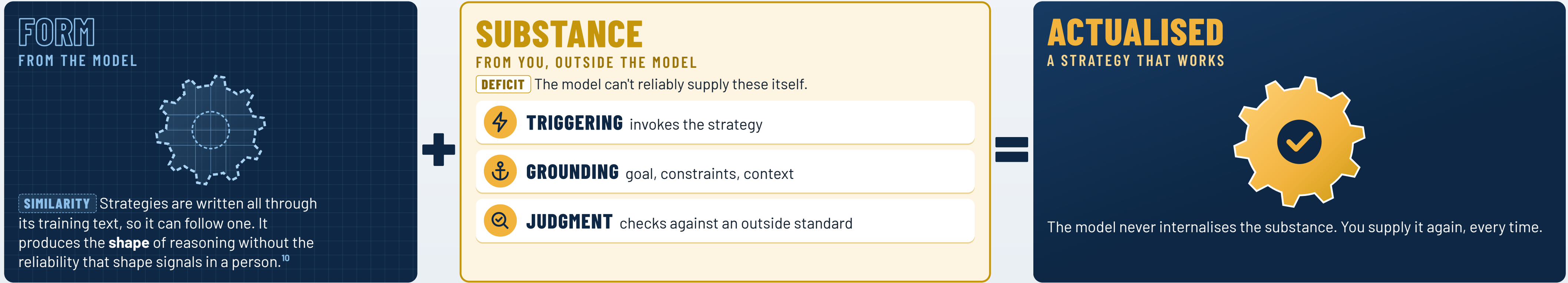
01 THE PATTERN Old strategies, new tool

The techniques that most reliably improve LLM output are strategies people already use to manage their own limits.

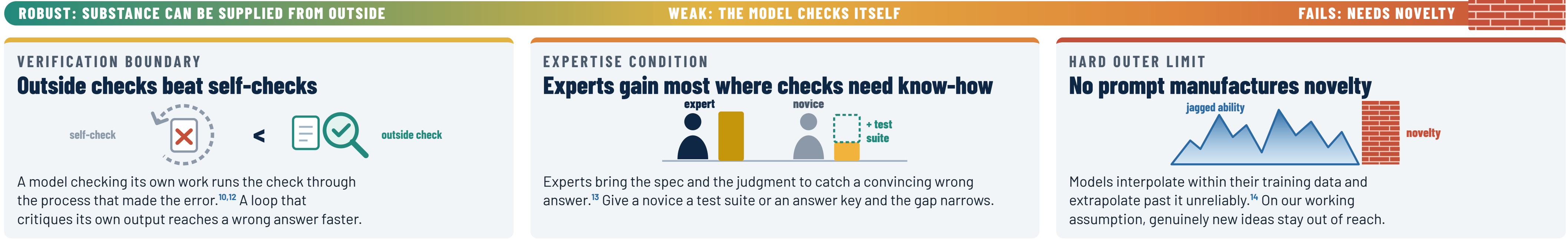


02 THE MECHANISM The model supplies the form. You supply the substance.

Two rival explanations, **SIMILARITY** and **DEFICIT**, turn out to be one mechanism seen from two ends.



03 THE BOUNDARIES Transfer holds as far as outside supply can reach



The skills that make AI work are already in your workforce.

- IS THEORY**
Explains why prompting works **without claiming the machine thinks**. Effective AI use is affordance actualisation.
- ORGANISATIONS**
Briefing, decomposing, specifying and reviewing already exist. **Pair AI with domain experts**. Fund the test suite before the prompt course.
- AI LITERACY**
Teach prompting as applied management, not incantations. Above all, teach **external verification**.
- NEXT**
Test the predictions: vary strategy and model capability, compare self-checks with outside checks, trial transfer-framed teaching with students.

ASK ME
Which of your existing skills is already an AI skill?
Scan for a worked example, a checklist and linked references
michael-borck.github.io/cst

AI USE Generative AI (Claude, Anthropic; Google Gemini) helped draft the poster text and create its graphics and layout, working from the author's manuscript. The research, framework and arguments are the author's own. **The author reviewed, verified and edited all content and takes full responsibility for it.**

¹ Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y. 2023).
² The Register (2025). Deloitte to refund Australian government over AI errors in report.
³ GPTZero (2026). Chasing the hallucinations: KPMG's AI-powered attempt at redefining excellence.
⁴ Markus & Silver (2008). A foundation for the study of IT effects. JAIS, 9(10/11).
⁵ Strong et al. (2014). A theory of organization-EHR affordance actualization. JAIS, 15(2).
⁶ Wei et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. NeurIPS.
⁷ Kojima et al. (2022). Large language models are zero-shot reasoners. NeurIPS.
⁸ Mollick (2026). Management as AI superpower. One Useful Thing.
⁹ Osmani (2025). How to write a good spec for AI agents.
¹⁰ Turpin et al. (2023). Language models don't always say what they think. NeurIPS.
¹¹ Cao et al. (2026). LLMs know when they know, but do not act on it. arXiv:2605.14186.
¹² Huang et al. (2024). Large language models cannot self-correct reasoning yet. ICLR.
¹³ Dell'Acqua et al. (2026). Navigating the jagged technological frontier. Organization Science, 37(2).
¹⁴ Zhao et al. (2025). Is chain-of-thought reasoning of LLMs a mirage? arXiv:2508.01191.